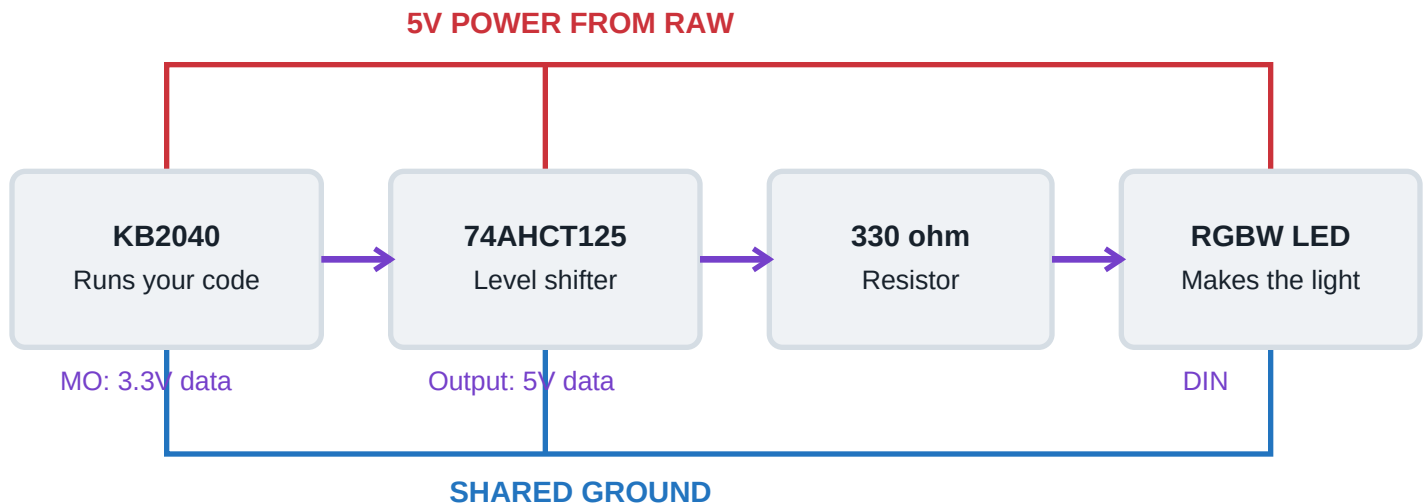


What we just built

A small circuit that lets your KB2040 control one colored LED.

Think of the KB2040 as the person speaking, and the level shifter as a speaker that repeats the message louder. The message is a color instruction. Power reaches the LED on a separate wire.



Power

RAW powers the chip and LED. The LED does not get its power through the resistor.

Ground

The chip, LED and knob share a path back to KB2040 G. Think of the perfboard as a shared junction.

Color instructions

MO sends data to the chip. The chip passes it through the resistor to the LED's DIN pad.

The other parts: the little **104 capacitor** smooths brief power disturbances. The remaining jumper wires keep the chip's three spare channels steady and switched off.

The perfboard is a holder with separate metal pads. **Adjacent holes are not connected until you join them.** Knob rotation and button signal wires are outside this LED diagram.

Reading the guide

2: full layout 3: data 4: power 5: ground 6: connection reference

Based on connections you reported; not a photo or electrical verification.

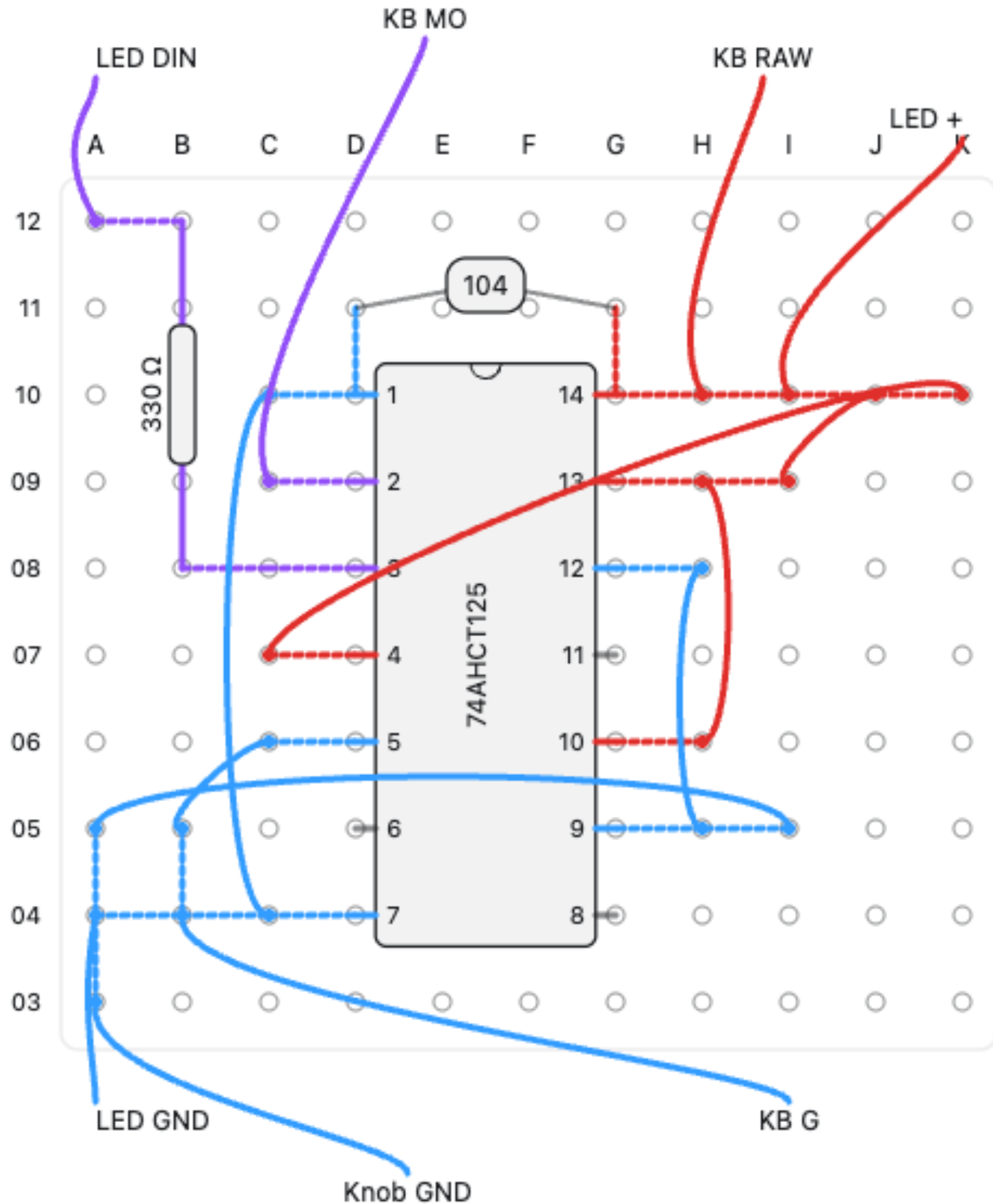
[KB2040 pinout](#)

[Chip datasheet](#)

[RGBW LED](#)

Your complete perfboard layout

Top view: letters across the top, chip notch facing up.

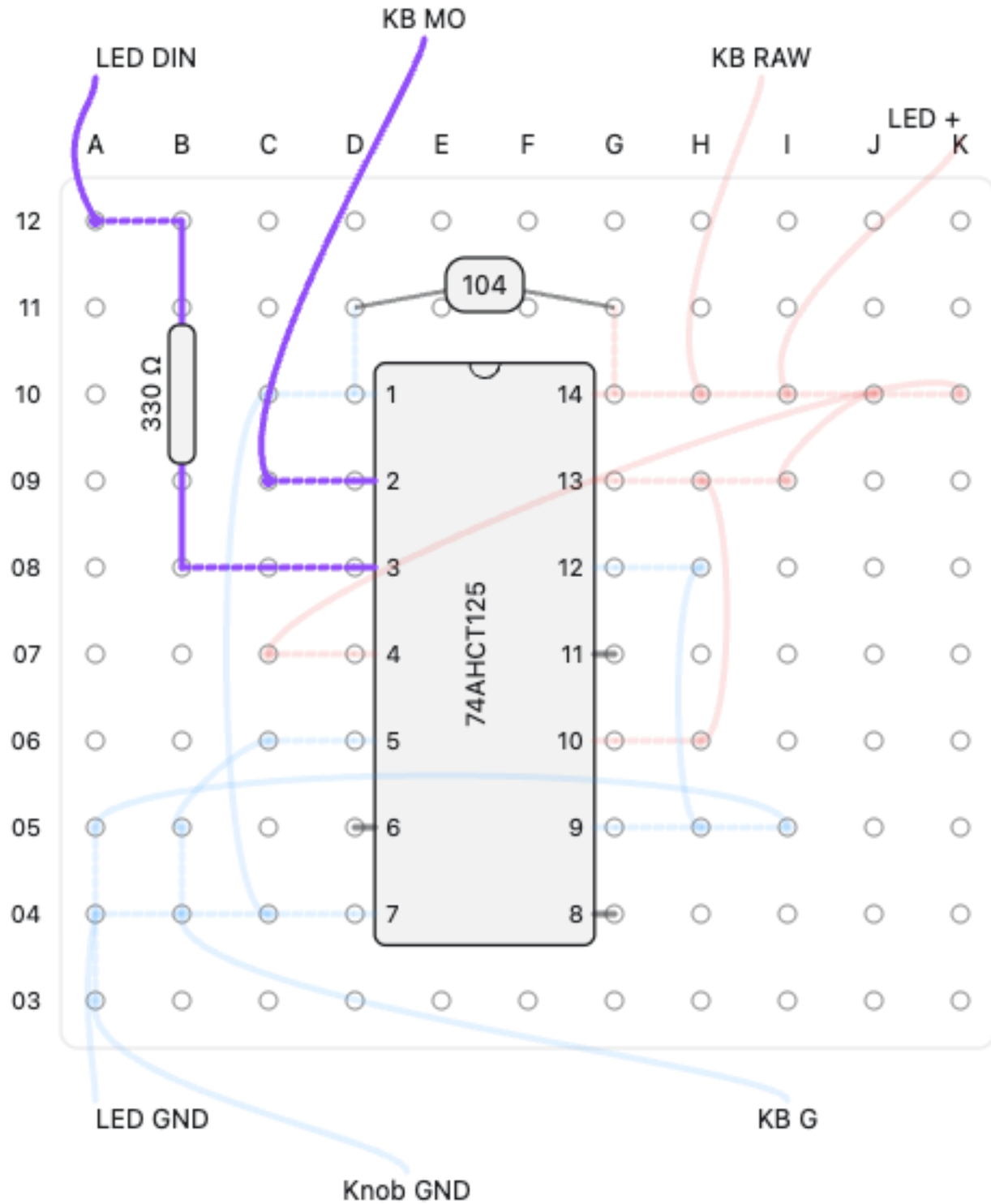


Solid lines: insulated wires on top. Dashed lines: bare solder links underneath.

Colors show function, not your wire insulation. Crossings alone do not connect.

Follow the color instructions

Purple is the data path. Other connections are faded.

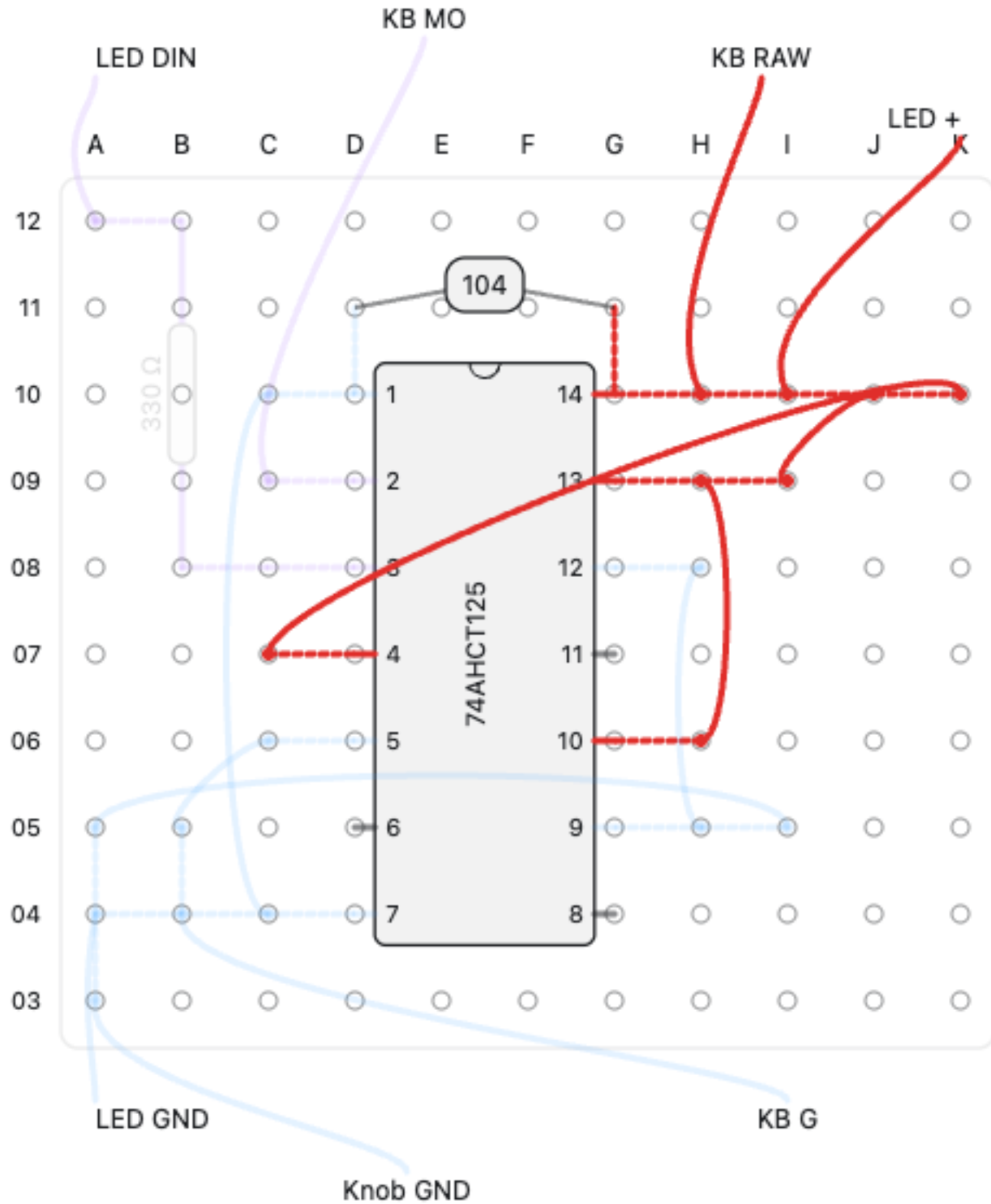


KB MO > C9-D9 > chip pin 2 > chip pin 3 at D8 > B8 > resistor > B12-A12 > DIN.

The resistor carries the signal; it is not the LED power supply.

Follow the power

Red is the RAW power network. Other connections are faded.

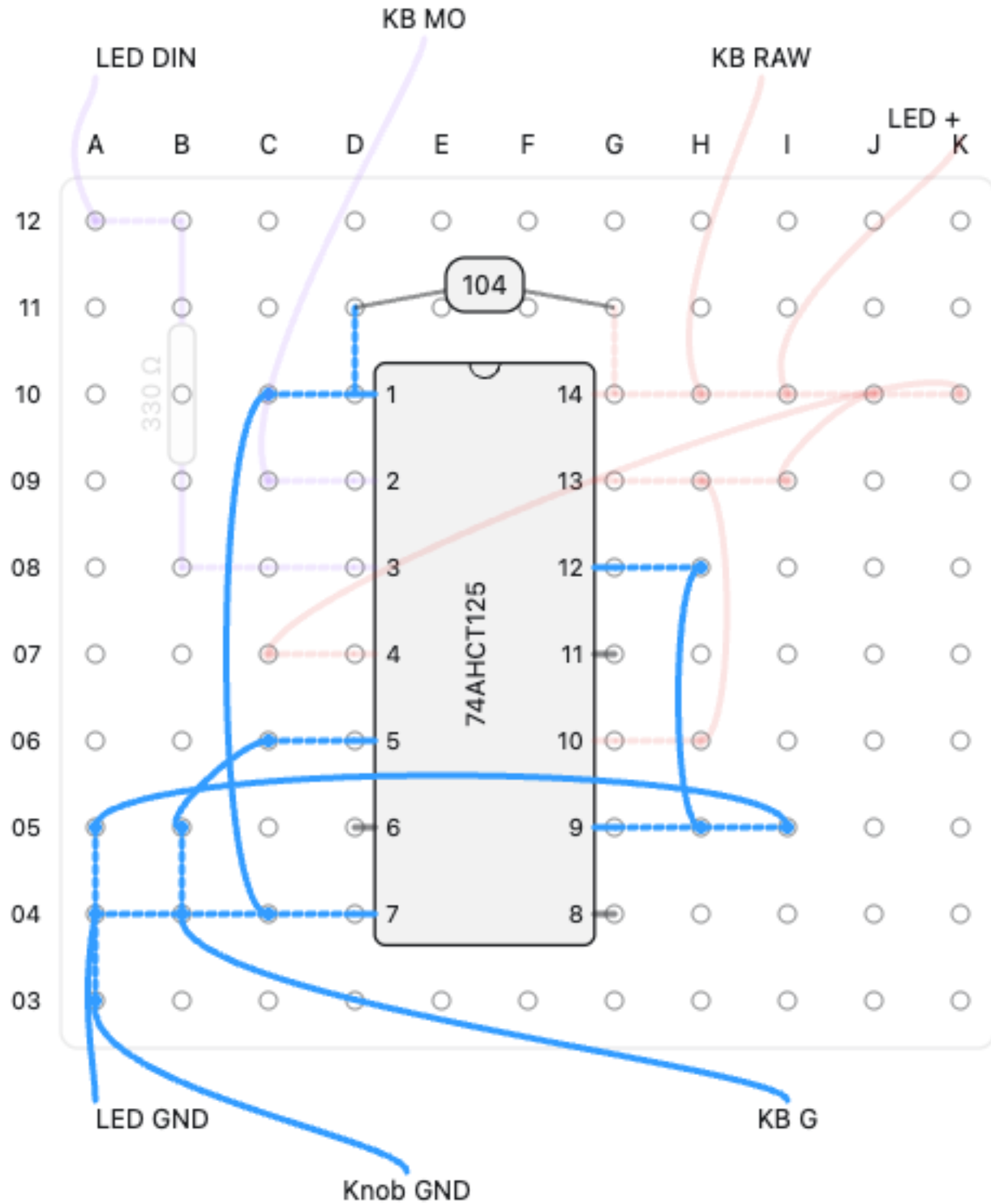


RAW > H10-G10 powers the chip; H10-I10 powers the LED.

The red jumpers to pins 4, 10 and 13 switch the three unused channels OFF.

Follow the shared ground

Blue connects the chip, LED and moved knob ground wire.



Knob > A3-A4 > B4 > KB G. LED GND also joins A4.

Grounding pin 1 turns our active channel ON; pins 5, 9 and 12 are unused inputs.

Hole-by-hole reference

Perfboard coordinates are always read from the TOP, even for underside joints.

Connection	Top-side endpoints	Join underneath
Active enable	C10 to C4	C10-D10; C4-D4
KB ground	B4 to KB G	B4-C4
KB power	H10 to KB RAW	H10-G10
Data input	C9 to KB MO	C9-D9
LED data	A12 to LED DIN	A12-B12
LED ground	A4 to LED GND	A4-B4
LED power	I10 to LED 5V	I10-H10
Spare input 1	C6 to B5	C6-D6; B5-B4
Spare inputs 2 + 3	H5 to H8	H5-G5; H8-G8
Ground that pair	I5 to A5	I5-H5; A5-A4
Disable pair	H6 to H9	H6-G6; H9-G9
Power that pair	I9 to J10	I9-H9; J10-I10
Last disable	C7 to K10	C7-D7; K10-J10
Moved knob ground	A3 to knob ground	A3-A4
104 capacitor	D11 to G11	D11-D10; G11-G10
330 ohm resistor	B8 to B12	B8-C8-D8; B12-A12

Leave these chip outputs unconnected: pin 6 at D5, pin 8 at G4, and pin 11 at G7.

Read a row like this: "C6 to B5" is one insulated wire across the top. Underneath, one bare end joins C6 to D6; the other joins B5 to B4.

Unplug USB before moving or soldering wires. An unlit LED alone does not identify a wiring fault.